

1	ACSB/2025/I/04
	(a) Write down and simplify the first four terms in the expansion of $\left(1 - \frac{x}{2}\right)^8$, in ascending powers of x. [2] (b) In the expansion of $(1 + ax + bx^2)\left(1 - \frac{x}{2}\right)^8$, the coefficients of x^2 and x^3 are -9 and 30 respectively. Find the value of each of the constants a and b. [4]
2	BASS/2025/I/05
	In the expansion of $\left(\frac{w}{x} + x\right)^8$, where w is a positive constant, the coefficient of $\frac{1}{x^2}$ and $\frac{1}{x^4}$ are equal. (a) Find the first 4 terms in the expansion of $\left(\frac{w}{x} + x\right)^8$ in ascending powers of x, simplifying each term. [4] (b) Find the value of w. [2] (c) Using the value of w, find the coefficient of x^{-4} in the expansion of $(x^2 - 5)\left(\frac{w}{x} + x\right)^8$. [3]
3	CCHY/2025/I/5
	(i) By considering the general term in the expansion of $\left(x^5 + \frac{1}{x}\right)^7$, explain why there are no even powers of x in its expansion. [2] (ii) Given that the coefficient of x^6 in the expansion of $\left(x^5 + \frac{1}{x}\right)^7 + (kx + 3)^7$ is 1344, where k is a positive constant, find the coefficient of x^4 in the expansion. [4]
4	GMS/2025/I/4
	Find the first four terms in the expansion of $\left(2x - \frac{p}{x^2}\right)^5$ in descending powers of x, simplifying each term. [3] (b) Given that the coefficient of x^{-1} in the expansion of $(4x^3 - 1)\left(2x - \frac{p}{x^2}\right)^5$ is $-160p^2$, find the value of p. [4]

5	FMSS/2025/I/5
	(a) Find the first 4 terms in the expansion of $\left(x^2 - \frac{2}{x}\right)^8$ in descending powers of x, simplifying each term. [2] (b) Given that the coefficient of x^{10} in the expansion of $(kx^3 + 5)\left(x^2 - \frac{2}{x}\right)^8$ is -336, find the value of k. [2] (c) Explain why there is no constant term in the expansion of $\left(x^2 - \frac{2}{x}\right)^8$. [2]

1	(a) $1 + {}^8C_1\left(-\frac{x}{2}\right) + {}^8C_2\left(-\frac{x}{2}\right)^2 + {}^8C_3\left(-\frac{x}{2}\right)^3 + \dots$ $1 - 4x + 7x^2 - 7x^3 + \dots$ (b) $b - 4a = -16$ $7a - 4b = 37$ $a = 3$ $b = -4$
2	5a $\left(x + \frac{w}{x}\right)^8 = w^8x^{-8} + 8w^7x^{-6} + 28w^6x^{-4} + 56w^5x^{-2} + \dots$ 5b $w^8x^{-8} + 8w^7x^{-6} + 28w^6x^{-4} + 56w^5x^{-2} + \dots$ Coefficient of $x^2 = 56w^5$ Coefficient of $x^4 = 28w^6$ $56w^5 = 28w^6$ $w = 2$ 5c $(x^2 - 5)\left(\frac{2}{x} + x\right)^8$ $(x^2 - 5)(\dots + 8(2)^7x^{-6} + 28(2)^6x^{-4} \dots)$ $= (x^2 - 5)(\dots + 1024x^{-6} + 1792x^{-4} \dots)$ Coefficient = $1024 + (-5)1792 = -7936$
3	(i) $T_{r+1} = \binom{7}{r} (x^5)^{7-r} \left(\frac{1}{x}\right)^r$ $= \binom{7}{r} (x)^{35-5r} x^{-r}$ $= \binom{7}{r} x^{35-6r}$ $6r$ is always even by all non-integers of r. $35 - 6r$ is always odd. $\therefore$ There are no even powers of x in its expansion. (ii)

	$T_{r+1} = \binom{7}{r} (kx)^{7-r} 3^r$ $= \binom{7}{r} (k)^{7-r} 3^r x^{7-r}$ $7-r=6$ $r=1$ $T_2 = \binom{7}{1} (k)^6 3^1 x^6$ $21k^6 = 1344$ $k = 2 \text{ or } k = -2 \text{ (rejected)}$ $7-r=4$ $r=3$ $T_4 = \binom{7}{3} (2)^4 3^3 x^4$ $\therefore \text{coefficient of } x^4 = 15120$		
4	(a) $\left(2x - \frac{p}{x^2}\right)^5 = (2x)^5 + 5(2x)^4 \left(-\frac{p}{x^2}\right) + 10(2x)^3 \left(-\frac{p}{x^2}\right)^2 + 10(2x)^2 \left(-\frac{p}{x^2}\right)^3 + \dots$ $= 32x^5 - 80px^2 + \frac{80p^2}{x} - \frac{40p^3}{x^4} + \dots$ (b) $(4x^3 - 1) \left(2x - \frac{p}{x^2}\right)^5 = (4x^3 - 1) \left(32x^5 - 80px^2 + \frac{80p^2}{x} - \frac{40p^3}{x^4} + \dots\right)$ Coefficient of x^{-1} $= 4(-40p^3) + (-1)(80p^2)$ $= -160p^3 - 80p^2$ $-160p^3 - 80p^2 = -160p^2$ $80p^2(2p - 1) = 0$ $p = 0 \text{ (Reject) or } p = 0.5$		
5	5(a)	$x^{16} - 16x^{13} + 112x^{10} - 448x^7$	5(b) $k = 2$
	5(c)	For term independent of x, $r = \frac{16}{3}$ Since r is not a positive integer, there is no term independent of x.	